IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM Granite and LangFlow"

Project Tile – AI Agent for Smart Farming Advice

Domain of Project – Agriculture

Faculty Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Ms. Jaina Patel		jainapatel.ce@ddu.ac.in	9726992794

Problem Statement - AI Agent for Smart Farming Advice

Problem Statement : AI Agent for Smart Farming Advice

The Challenge

Farmers often face difficulties in making informed decisions regarding crop selection, irrigation scheduling, pest management, and market planning. Agricultural knowledge is scattered across multiple sources, making it difficult to access timely and reliable recommendations.

The Objective

Build an AI-powered Smart Farming Advisor using LangFlow and IBM Granite models that provides crop guidance, weather-based irrigation advice, pest diagnosis, and market insights through a collaborative multi-agent system.

Technology Used

I used

- LangFlow
- IBM watsonx.ai
- IBM Granite Models
- Multi-Agent Architecture

Proposed Solution -

- The proposed Smart Farming AI is a multi-agent agricultural advisory system built using LangFlow and IBM Granite models. Four specialized agents handle crop advisory, weather & irrigation planning, pest and disease diagnosis, and market intelligence. A Supervisor Agent combines their outputs into a single farmer-friendly recommendation report.

- The solution improves farming productivity, reduces risks, and supports better decision-making.

Langflow component Used -

1) Chat Input – Interface where farmers enter crop details, soil information, weather conditions, and farming issues through natural language queries.

2) Crop Advisory Agent (IBM watsonx AI Agent) – Uses IBM Granite models to analyze crop and soil information and provide seed recommendations, fertilizer guidance, harvest timelines, and yield improvement suggestions.

3) Weather Advisory Agent (IBM watsonx AI Agent) – Evaluates weather-related inputs to recommend irrigation schedules, identify weather risks, and suggest preventive farming actions.

4) Pest & Disease Agent (IBM watsonx AI Agent) – Analyzes crop symptoms to identify possible pests or diseases and provides treatment and prevention recommendations.

5) Market Insights Agent (IBM watsonx AI Agent) – Generates market demand insights, selling strategies, cost optimization suggestions, and profit improvement recommendations.

6) Supervisor Agent (IBM watsonx AI Agent) – Combines outputs from all specialized agents into a single comprehensive farming advisory report with actionable recommendations.

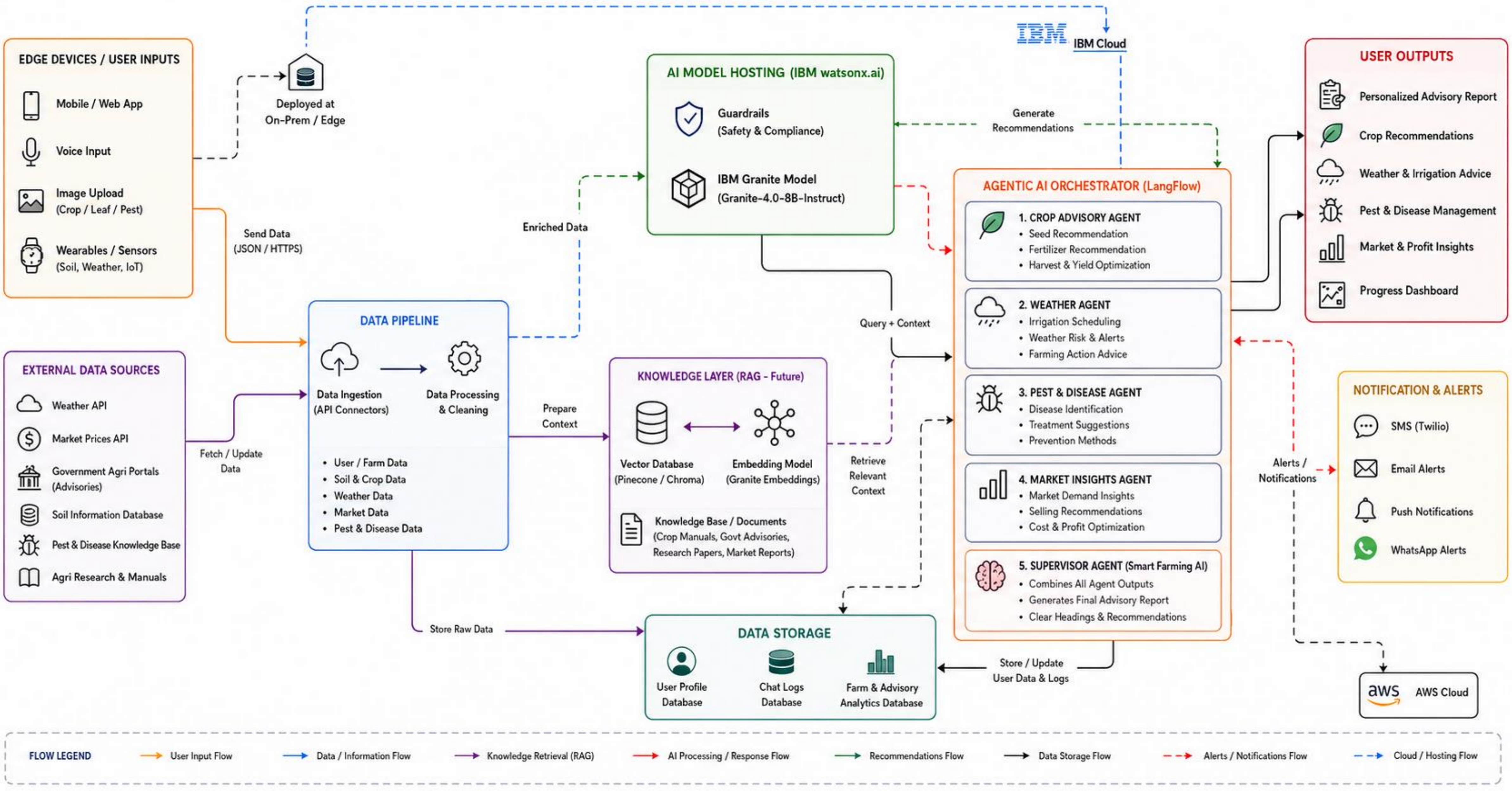
7) Name of IBM Granite Model – **Granite-4.0-8B-Instruct** → Fast and efficient model for real-time agricultural advisory and multi-agent decision support.

8) Chat Output – Displays the final consolidated farming report including crop advice, weather recommendations, pest management guidance, market insights, and overall farming recommendations.

This matches the style and level of detail of your nutrition-agent sample slide

Langflow workflow -

Architecture blueprint



Role of Agentic AI in the solution

Agentic AI enables multiple specialized agents to work collaboratively.

- Crop Agent – seed, fertilizer, yield advice
- Weather Agent – irrigation and weather-risk planning
- Pest Agent – disease diagnosis and treatment
- Market Agent – pricing and profitability insights
- Supervisor Agent – combines outputs into one final recommendation

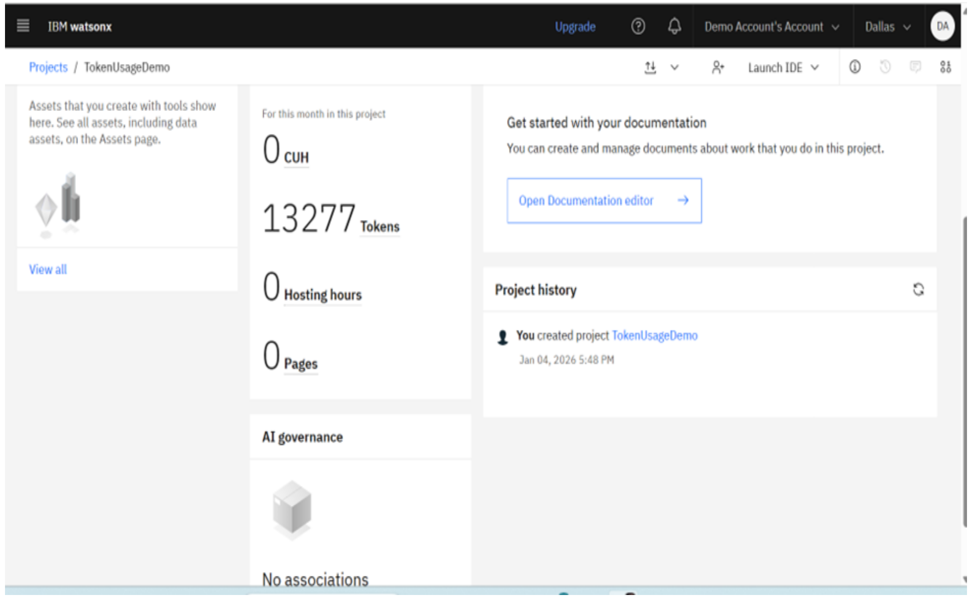
This creates an intelligent, context-aware farming assistant.

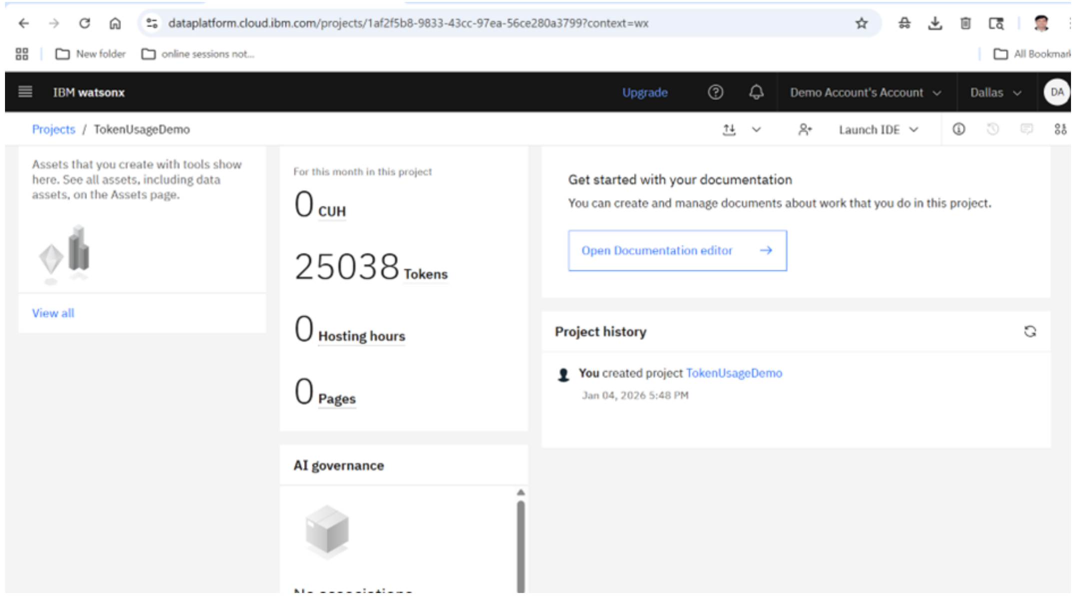
Technology Used

- LangFlow – Workflow orchestration
- IBM Granite 4.0 8B Instruct – Reasoning and recommendations
- IBM watsonx.ai – Model hosting and AI services
- Multi-Agent Architecture – Specialized advisory system

Token Usage

11,761 token are used





Novelty and Uniqueness

- Multi-Agent Agricultural Advisory System
- Specialized expert agents for farming tasks
- Supervisor-based decision aggregation
- Scalable architecture for future RAG integration
- Real-world agricultural use case
- Built using IBM Granite Models and LangFlow

Git Hub Link

Github URL – https://github.com/jainapatelce/BVM_FDP

-
-
-

Future Scope

1. RAG Integration with government advisories, crop manuals, and agricultural datasets.

2. Real-time weather APIs and satellite data integration.

3. Multilingual farmer support.

4. Mobile application deployment.

5. Image-based disease detection using computer vision.

.

.

.

Thank You!

Thank you for your time and interest.